

Bhuvanesh Srikanth

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Profile

Computer Science undergraduate with a strong foundation in full-stack development and systems design. Proven track record of shipping complete, production-quality applications. Passionate about building software that is both technically rigorous and genuinely useful to the people who use it.

Education

Amrita Vishwa Vidyapeetham

Aug 2023 – Late 2027

- CGPA: 7.53/10
- Coursework: DSA, ML, OS, DBMS, Computer Networks, OOP, Edge Computing, Full Stack Frameworks

Experience

Organizing Committee Member — *IGNITE '24, Leadership & Innovation Summit*

2024

- Selected as a core team member; coordinated speaker logistics, hospitality, and event operations for hundreds of attendees across a multi-day premier campus summit.

Projects

GrowingPlate — *Children's Nutrition Monitoring Platform*

growing-plate.vercel.app 

- Architected a full-stack pediatric nutrition tracker with age- and sex-specific RDA targets (ICMR 2020 / WHO) for children; designed a relational PostgreSQL schema with Row Level Security enforcing strict per-parent data isolation.
- Built a zero-dependency pure TypeScript nutrition engine handling gram conversion, per-item nutrient computation, and a four-tier severity alert system (green/yellow/orange/red) against RDA targets.
- **Stack:** Next.js 14 (App Router, TypeScript), Supabase (PostgreSQL + RLS), Tailwind CSS, shadcn/ui, Recharts, Framer Motion, Zod

Timetable Scheduler — *Academic Scheduling Platform*

[GitHub Repository](#) 

- Implemented a constraint-satisfaction backtracking engine generating clash-free university timetables, automatically resolving faculty workloads, room capacities, consecutive labs, and parallel elective conflicts.
- Delivered tri-tier RBAC (Admin/HOD/Faculty) with JWT auth, SMTP schedule notifications, semester rollover, audit logging, and CSV import/export via a full-featured admin dashboard.
- **Stack:** Django / DRF, PostgreSQL, React + Vite, pytest with coverage, CI/CD via GitHub Actions

Edge-Based Smart Braking System

[GitHub Repository](#) 

- Built a dual-layer collision avoidance system on an ESP32-CAM: an ultrasonic sensor triggers immediate plug-braking when obstacles are detected within a configurable threshold (default 20cm), while a cloud-hosted CNN (FastAPI on Hugging Face Spaces) classifies live camera frames to output steering decisions.
- Implemented a fuzzy logic distance controller with sigmoid-based membership functions across four proximity zones (very close/close/far/very far), producing a continuous action score that smoothly transitions between STOP, SLOW DOWN, and GO rather than relying on brittle hard thresholds.
- **Stack:** C++ (Arduino/ESP32), Python, FastAPI, YOLOv8, PyTorch, ThingSpeak

Technologies

Languages: Python, JavaScript, TypeScript, C, C++, SQL, HTML, CSS

Frameworks & Tools: React, Next.js, Node.js, Express.js, Django, FastAPI, PostgreSQL, MongoDB, Supabase, Docker, Git, Vercel, Web3.js

Concepts: REST API Design, Relational DB Design, Row Level Security, Constraint Satisfaction, NLP, Graph Databases, Edge Computing, JWT Auth, CI/CD